Automated Feature Extraction RFI Objective Questions

Utilizing Section 3.0, 4.0, and Appendix A of the FDT AFE SOO as a reference, describe how you would address the following:

Objective 1: Provide and Deploy a High-Performing Automated Feature Extraction (AFE) Capability or Service:

What AFE automation processes do you utilize to produce features today? For a given source (or combination of sources), what is your methodology to create real world representations of specific features? What combinations of data types are used in your process (e.g. vector-to-vector, image-to-vector, PDF-to-vector, and/or image-to-image, image-to-surface model, etc.) for specific features? How do your current processes ensure accurate position, confidence assessment and time is associated with automated feature extraction processing and its final product? How would attribution and identification of source data be handled throughout the AFE processing chain? Can you describe how your AFE solution(s) fits into geospatial data production tools and systems utilizing multiple input data formats? How do your capabilities integrate with other commercially available geospatial tools? What Technical Readiness Level[1] (TRL) is your capability current graded at? Describe the process on how you would separate out multiple features that may be conflated during AFE processing.

Objective 2: Ensure High-Quality Data and Enable Analyst-in-the-Loop Validation:

How would you ensure that the changes that may occur during the feature extraction processing are detected and retained are relevant to geospatial data? What source formats would your system or algorithm(s) ingest to produce automated specific and/or multiple features within the same area of interest? What processes would you employ to provide GEOINT assurance that changes are verified, comprehensive and valid? Describe your process to organize an analyst notification system and how would you ensure critical processes/outputs are prioritized for review? What are your defined threshold metrics for minor, major and critical for AFE outputs for topographic, maritime and aeronautical features? How would you achieve an objective threshold accuracy of 80% with a goal for objective accuracy of 99% for topographic, maritime and aeronautical features? How does your solution interface with GIS systems for viewing, evaluating and editing results? Describe any AI/ML models that are used for your quality assessments.

Objective 3: Enable Seamless Utilization by NGA's GEOINT Ecosystem via open APIs

Describe the APIs your solution provides to allow external system (like NGA's) to programmatically submit processing jobs, monitor status, and retrieve results. Please specify the API types (e.g., REST, gRPC), authentication methods, and data formats used. What standardized formats (e.g., GDB, Shapefile) to support interoperability with other NGA systems and direct dissemination to customer platforms would you recommend? What would your compute and infrastructure needs be for your solution?

Objective 4: Meet and Maintain Stringent Performance and Security Standards

Describe your solution's scalability, reliability, resiliency and security standards used in the production of AFE products. What are the nominal/baseline operating benchmarks and infrastructure specifications for your system?
